

Iniyen Andrews Joseph

Doctoral Applicant in Computer Science | Aalto University & ELLIS Institute Finland

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Location: Espoo / Helsinki Region, Finland | Primary Call: Autumn 2026 ELLIS Doctoral Programme

RESEARCH INTERESTS

Stochastic Control & Deep Reinforcement Learning (PPO), Generative World Models & Diffusion Processes, High-Dimensional Non-Stationary State-Space Modeling, 3D Kinematics & Biomechanical Digital Twins, and Multimodal Neuro-Symbolic Foundation Architectures.

EDUCATION

National Institute of Electronics & Information Technology (NIELIT), Calicut, India

Master of Science in Computer Science (Specialization: Data Science)

Expected May 2026

CGPA: 8.55 / 10.0

- ▶ **M.Sc. Thesis:** *Reinforcement Learning for Autonomous Racing in Simulated Continuous Environments (Trackmania)*
- ▶ **Supervisor:** Dr. Vimala Mathew (Scientist E, Department of IT, NIELIT Calicut)
- ▶ **Methodology & Focus:** Formulated high-speed trajectory planning as a continuous action-space Markov Decision Process (MDP). Engineered policy gradient algorithms (PPO) with dynamic reward shaping to navigate non-linear contact friction, high-speed inertial drift, and latency-bounded state estimation.

The American College, Madurai, India

Bachelor of Science in Computer Science

2021 – 2024

CGPA: 7.4 / 10.0

RESEARCH & TECHNICAL PROJECTS

Procedural 15-DOF Kinematic Hand Rig & ASL-LEX Phonological Compiler

2025 – 2026

Lead Architect & Researcher | Three.js, WebGL, CCD-IK

[\[Live Demo\]](#) | [\[Proposal PDF\]](#)

- ▶ **Biomechanical 15-DOF Kinematic Core:** Engineered an in-browser, 60 FPS 15-DOF anatomical hand kinematics engine mapping discrete linguistic phonological morphemes into continuous 3D joint rotations for continuous Sign Language Production (SLP).
- ▶ **Anatomically Clamped CCD-IK Solver:** Implemented a 3-iteration CCD Inverse Kinematics solver incorporating Kapandji anatomical joint limits (MCP flexion $[-10^\circ, 90^\circ]$ / abduction $[-15^\circ, 15^\circ]$, PIP $[0^\circ, 100^\circ]$, DIP $[0^\circ, 80^\circ]$) with rest-pose spring biasing for lateral digits and a 5-DOF saddle thumb.
- ▶ **Dynamic Obstacle & Collision Avoidance:** Built collision-avoidance temporal priority queues for folded/tucked signs (M, N, T) and Catmull-Rom spline trajectory generation for dynamic strokes (J, Z), eliminating kinematic self-intersection.
- ▶ **Empirical Phonological Validation:** Evaluated across 2,723 lexical signs from ASL-LEX 2.0, providing empirical validation for neuro-symbolic procedural scaffolding to bound downstream residual diffusion models.

Deep Reinforcement Learning for Non-Stationary Stochastic Control (Project Bithax)

2025 – 2026

Lead Researcher | Python, PyTorch, Gymnasium, Stable-Baselines3, PPO

[\[View on GitHub: bithax\]](#)

- ▶ **Continuous State-Space Formulation:** Formulated high-frequency continuous cryptocurrency market dynamics (BTC/USD) as a continuous state-space Markov Decision Process (MDP) over a 42-dimensional feature space in non-stationary stochastic environments.
- ▶ **Risk-Sensitive Multi-Objective Reward:** Engineered reward functions incorporating a 0.05% transaction friction penalty and volatility-penalized variance terms to resolve the multi-objective policy trade-off between execution frequency and risk-adjusted return.
- ▶ **Empirical Generalization & Zero Look-Ahead Bias:** Conducted rigorous out-of-sample walk-forward stress testing and look-ahead bias elimination, achieving a Sharpe Ratio of ≈ 1.4 , Profit Factor of 1.9, and Win Rate of 76.6% against standard buy-and-hold benchmarks.

RESEARCH & TECHNICAL PROJECTS (CONTINUED)

Generative World Models for High-Frequency Stochastic Microstructure 2026
Lead Researcher | Python, PyTorch, Transformers, Generative Modeling [View on GitHub: Generative_LOB_Simulator]

- ▶ **Autoregressive Microstructure Modeling:** Developed autoregressive generative sequence architectures to model the high-dimensional, asynchronous event dynamics of Limit Order Books (LOB).
- ▶ **Non-Markovian Order Flow Simulation:** Investigated transformer-based world models to capture non-Markovian queue depletion, order-flow transitions, and fat-tailed price impact distributions in financial microstructure, establishing a computational framework for generative simulation.

Variance Reduction in High-Noise Regimes: Deep Learning vs. Regularized Baselines (Project HOPE) 2024 – 2025
Independent Researcher | Scikit-Learn, TensorFlow, LSTM Networks [View on GitHub: HOPE]

- ▶ **Low-SNR Generalization Benchmark:** Conducted a systematic empirical study comparing deep recurrent neural networks (LSTMs) against regularized linear state estimators for daily directional return prediction across the S&P 500 and Nifty 50 indices.
- ▶ **Variance-Constrained Optimization:** Demonstrated that in low signal-to-noise ratio (SNR) stochastic regimes, strongly regularized linear models outperform over-parameterized deep neural networks by constraining estimator variance, attaining an annualized Sharpe Ratio > 2.5 and 70.6% win rate.

Jaguu: Multimodal Edge Foundation Agent 2025
Lead Developer | Llama 3, Ollama, PyTorch, Computer Vision [View on GitHub: Jaguu]

- ▶ **Local Foundation Agent Architecture:** Architected a privacy-first, local multimodal voice and reasoning agent coupling quantized autoregressive LLMs (Llama 3) with computer vision backbones.
- ▶ **Sub-Second Edge Token Inference:** Engineered low-latency cross-modal token conditioning pipelines for compute-constrained edge deployment, achieving sub-second token response on consumer-grade hardware.

TECHNICAL & MATHEMATICAL COMPETENCIES

Stochastic Control & RL:	Markov Decision Processes (MDPs), Proximal Policy Optimization (PPO), Deep Q-Networks (DQN), Risk-Sensitive Reward Engineering, Stable-Baselines3, Gymnasium.
3D Kinematics & Graphics:	Cyclic Coordinate Descent (CCD) IK, Analytical 2-Bone Arm IK, Lie Algebra ($\mathfrak{se}(3)$), Quaternions, Euler Transforms, Three.js, WebGL, Biomechanical Hand/Arm Rigs, Blender3D.
Deep Learning & Foundation AI:	PyTorch, Hugging Face Transformers, Autoregressive LLMs (Llama 3), Diffusion Models, Fully Sharded Data Parallel (FSDP), WebDataset, Distributed Training.
Systems & HPC Infrastructure:	Python (Advanced), C++, JavaScript/ES6, SQL, Linux, Git, Docker, $\LaTeX$, AMD ROCm / HIP Environments (CSC LUMI Supercomputer Readiness).

SELECTED GRADUATE COURSEWORK & ACADEMIC MERIT

Relevant Graduate Coursework (NIELIT Calicut): Advanced Machine Learning, Deep Learning Architectures, Reinforcement Learning, Stochastic Processes & Probability, Statistical Data Modeling, Advanced Algorithms, Data Structures in C++.

Academic Standing: First Class with Distinction, M.Sc. Computer Science (CGPA: 8.55 / 10.0).

ACADEMIC REFERENCES

Dr. Marco Marini Research Mentor (SLT/SLP Formalization) Sapienza University of Rome Rome, Italy Email: marco.marini@uniroma1.it	Dr. Vimala Mathew Scientist E & M.Sc. Thesis Supervisor Department of Information Technology National Institute of Electronics & IT (NIELIT) Calicut, Kerala, India Email: vimala@calicut.nielit.in
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